

KATHERINE LALLOTIS

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EDUCATION

The Ohio State University

Columbus, OH

- Doctorate of Philosophy (PhD), Physics; Adviser: Christopher M. Hirata
Thesis: Optimizing Joint Processing for Stage IV Weak Gravitational Lensing Surveys Expected 2026
- Master's of Science (MS), Physics; Adviser: Christopher M. Hirata December 2023

Whitman College

Walla Walla, WA

- Bachelor of Arts in Physics & Astronomy, Minor in Mathematics May 2021
Honors: Magna Cum Laude, Honors in Major, Walter A. Brattain Scholarship

HONORS AND AWARDS

- Dept. of Energy Office of Science Graduate Student Research Fellowship 2024-2025
- Physics Service Courses Outstanding Graduate Teaching Assistant Award 2024
- Physics Departmental Service Award, for service on Climate and Diversity Committee 2024
- Honorable Mention, NSF Graduate Research Fellowships Program 2022
- University Fellowship, the Ohio State University 2021-2022

FIRST AUTHOR PUBLICATIONS

- **Lalotis, Katherine**, et al. "Analysis of Biasing from Noise from the Nancy Grace Roman Space Telescope: Implications for Weak Lensing." Publications of the Astronomical Society of the Pacific, vol. 136, no. 12, 1 Dec. 2024, p. 124506, <https://doi.org/10.1088/1538-3873/ad9bec>.
- **Lalotis, Katherine**, Burt, J. A., Mamajek, E. E., Li, Z., Perdelwitz, V., et al. (2023). Doppler constraints on planetary companions to nearby sun-like stars: An archival radial velocity survey of southern targets for proposed NASA Direct Imaging Missions. The Astronomical Journal, 165(4), 176. <https://doi.org/10.3847/1538-3881/acc067>

CO-AUTHORED PUBLICATIONS

- Cao, K., Hirata, C., **Lalotis, K.**, Yamamoto, M., Macbeth, E., & Troxel, M. (2025). Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. IV. Hyperparameter Optimization and Experimental Features. arXiv e-prints, arXiv:2509.18286.
- OpenUniverse, et al. "OpenUniverse2024: A Shared, Simulated View of the Sky for the next Generation of Cosmological Surveys." ArXiv.org, 2024, arxiv.org/abs/2501.05632.
- Cao, K., Hirata, C., **Lalotis, K.**, Yamamoto, M., Macbeth, E., & Troxel, M. (2025). Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. III. Software Improvements and New Linear Algebra Strategies. APJS, 277(2), 55.

- Yamamoto, M., **Laliotis, K.**, Macbeth, E., Zhang, T., Hirata, C., et al. (2024). Simulating image coaddition with the Nancy Grace Roman Space Telescope - II. Analysis of the simulated images and implications for weak lensing. MNRAS, 528(4), 6680-6705.
- Hirata, C., Yamamoto, M., **Laliotis, K.**, Macbeth, E., Troxel, M., et al. (2024). Simulating image coaddition with the Nancy Grace Roman Space Telescope - I. Simulation methodology and general results. MNRAS, 528(2), 2533-2561.

PUBLIC WRITING

- **Laliotis, Katherine.** “Europe’s Extremely Large Telescope Faces a New Dire Threat.” Scientific American, 30 Jan. 2025, www.scientificamerican.com/article/europes-extremely-large-telescope-faces-a-new-dire-threat/.
- **Laliotis, Katherine.** “Saving the Chandra X-Ray Observatory.” Undark Magazine, 19 Sept. 2024, undark.org/2024/09/19/opinion-saving-chandra-x-ray-observatory/.

PRESENTATIONS & POSTERS

Conferences & Collaborations

- NASA Physics of the Cosmos Early Career Workshop Sept. 2025
Presentation: Characterizing and Mitigating Noise on Roman Space Telescope Detectors for Precise Weak Gravitational Lensing
- Cosmic Cartography with Roman Conference, STScI July 2025
Poster: Removing correlated noise from the Roman sensor chip array through destriping algorithms
- Roman High Latitude Imaging Survey (HLIS) PIT In-Person Meeting, IPAC/CalTech Oct. 2024
Presentation: Analysis of biasing from noise on the Nancy Grace Roman Space Telescope and Implications for Weak Lensing
- Image Sensors for Precision Astronomy, SLAC March 2024
Poster: Read Noise Biasing on the Nancy Grace Roman Space Telescope

Public Engagement

- Speaker Series, San Mateo County Astronomical Society Oct. 2025
Lecture: Illuminating the Universe Through Weak Gravitational Lensing

TEACHING AND MENTORSHIP

- Mentored and co-advised undergraduate researchers in the Hirata group at OSU:
Emily Macbeth — August 2023 - May 2025
Amy Albert — August 2025 - Present
- Academic Facilitator: Undergraduate Residential Summer Access (URSA) Program, Summer 2024
Designed and taught a 2-week group-work centered curriculum exploring the question “What is Time?” through the lens of astrophysics for incoming OSU physics and astronomy students from underrepresented backgrounds in STEM.
- Teaching Assistant, OSU Physics, Fall 2022 & Spring 2023
Taught two sections of 30 students in twice weekly recitations and weekly labs; held twice-weekly office hours for students; and designed study guides and practice problems for students before exams.

LEADERSHIP & OUTREACH

Roman HLIS Project Infrastructure Team

Co-Chair, Junior Members of the PIT (JuMP)

May 2025 - Present

- Created the HLIS PIT Mentorship program, and launched the first cohort
- Organized 2 in-person JuMP events
- Organized 2 virtual JuMP events, including initiating a regular lecture series

News Coordinator

Sept. 2024 - Present

- Manage the News page on the PIT website; solicited 2 articles from contributing members and wrote 3 articles

Polaris Program, the Ohio State University

April 2022 - November 2024

Parliamentarian

- Planned and facilitated weekly meetings of the 12-person Polaris organization Leadership Team of grads and undergrads across different subfields of physics and astronomy
- Created and documented new Recruitment & Marketing, Grants, and High School Outreach committees
- Presented a twice-yearly report to community stakeholders detailing the use of our \$60,000 budget, the successes of our outreach programs, and ideal areas of prospective growth

URSA Polaris Program, the Ohio State University

General Facilitator

Summer 2022

- Marketed URSA to over 100 incoming physics/astronomy students from underrepresented backgrounds via mail, email, phone, and social media
- Organized logistics of the 15-student program, including housing, meals, transportation, technology access, classroom spaces, and communication channels between facilitators and participants
- Designed and taught a 2-week engaging, group-work centered curriculum exploring the question “*What is Time?*” through the lens of astrophysics